

Exam Date & Time: 18-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: ECE 2122/ECE_2122 NETWORK ANALYSIS

(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

NETWORK ANALYSIS [ECE 2122]

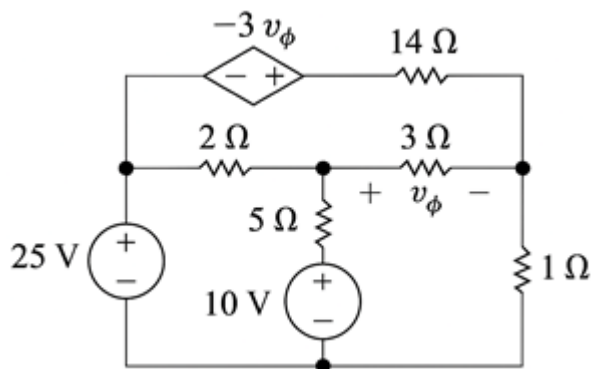
Marks: 50

Duration: 180 mins.

A

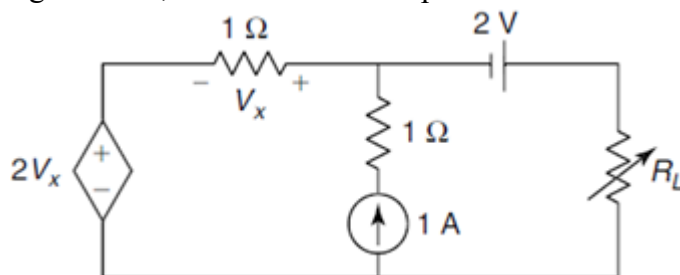
Answer all the questions.

- 1A) Use the mesh-current analysis to obtain the power delivered by the dependent voltage source.



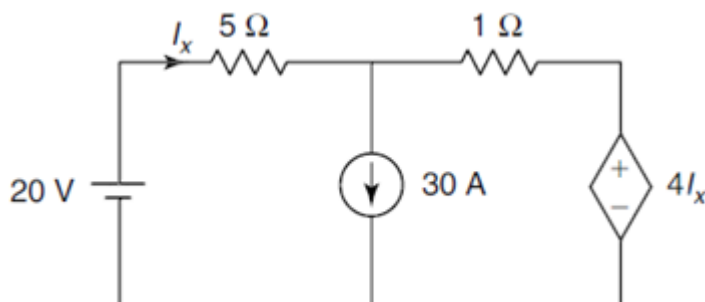
(4)

- 1B) Compute the value of R_L for maximum power transfer for the circuit shown in the figure. Also, find the maximum power.



(4)

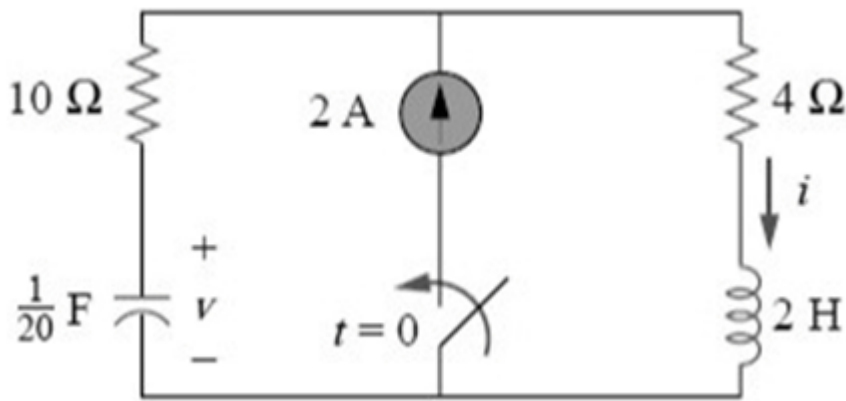
- 1C) Determine the current I_x in the circuit shown using the superposition theorem.



(2)

- 2A) Obtain the complete response v and then i for $t > 0$ in the circuit shown in Figure (Assume circuit is initially unenergized).

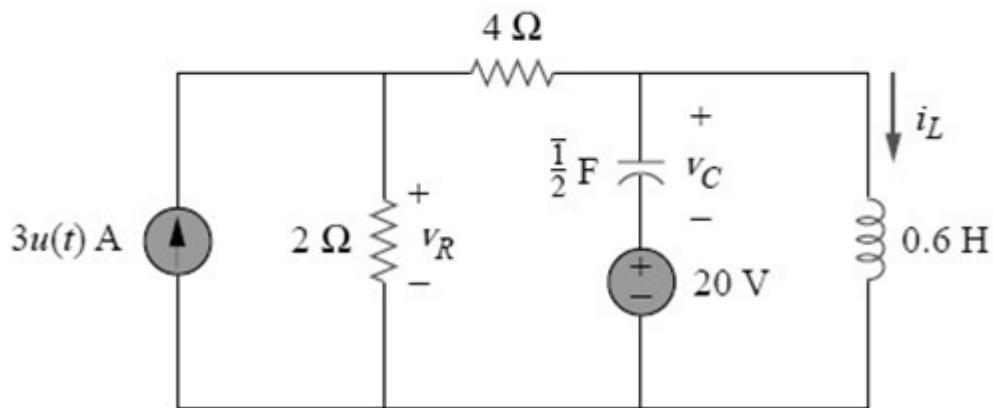
(4)



2B)

In the network shown in Figure, calculate

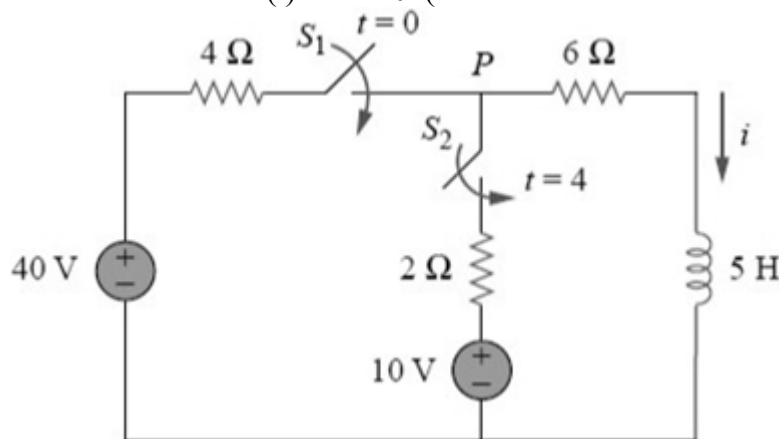
- (a) $i_L(o^+)$, $v_C(o^+)$, $v_R(o^+)$
- (b) $\frac{di_L}{dt}(o^+)$, $\frac{dv_C}{dt}(o^+)$, $\frac{dv_R}{dt}(o^+)$
- (c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$.



(4)

2C)

In the network shown in Figure, at $t = 0$, switch 1 is closed, and switch 2 is closed 4 seconds later. Find $i(t)$ for $t > 0$. (Assume inductor has zero initial current)

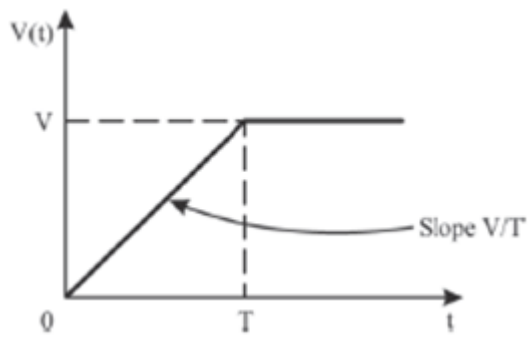


(2)

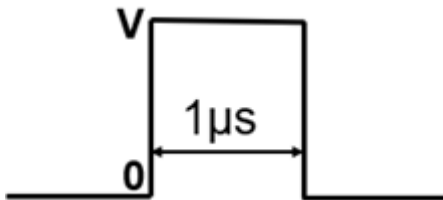
3A)

The limited ramp shown in Figure is applied to a RC differentiator. Draw the output waveform for (a) $T=0.2 RC$, (b) $T=RC$, (c) $T=5RC$.

(4)

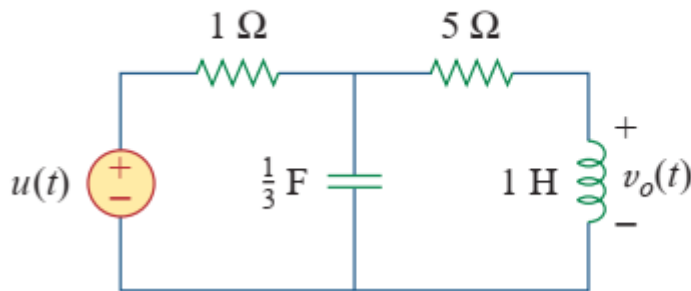


- 3B) An ideal $1\mu\text{s}$ pulse is fed to an amplifier. Calculate and plot the output waveform under the following conditions. The upper 3-dB frequency is (a) 8 MHz (b) 2 MHz.

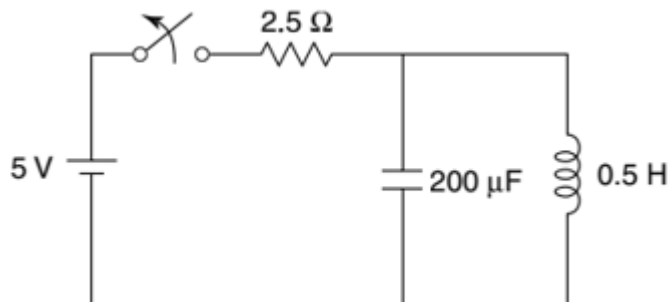


- 3C) Describe the concept of transmission error in a low-pass RC circuit when a ramp input signal is applied. (2)

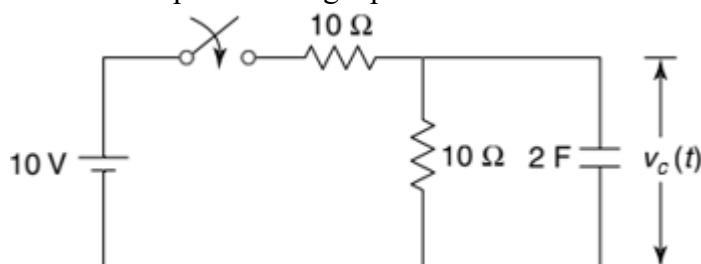
- 4A) Find $v_o(t)$ in the circuit, assuming zero initial conditions using laplace transform.



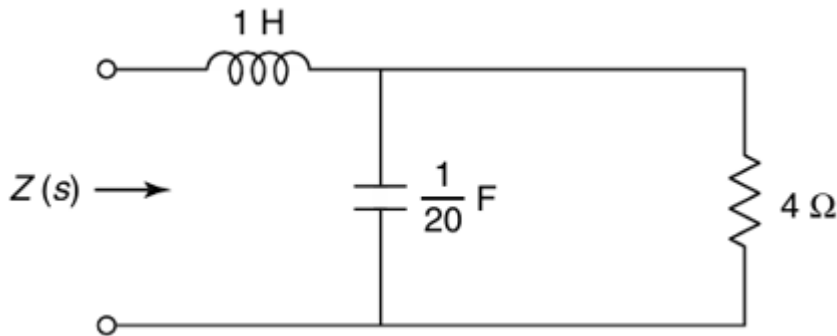
- 4B) In the network of Figure, the switch is closed and a steady state is attained. At $t = 0$, the switch is opened. Determine the current through the inductor using laplace transform.



- 4C) The switch in the network shown in the Figure is closed at $t = 0$. Determine the voltage across the capacitor using laplace transform.

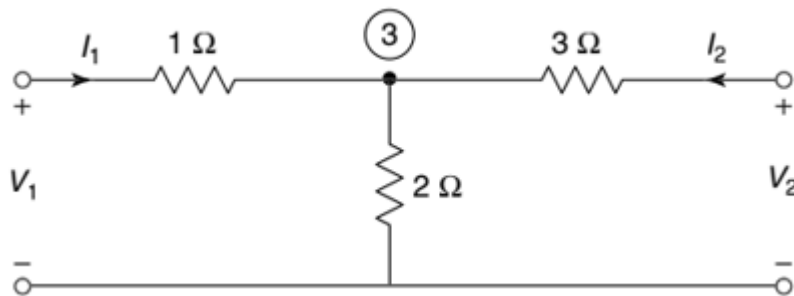


5A) Determine $Z(s)$ in the network shown in Figure.



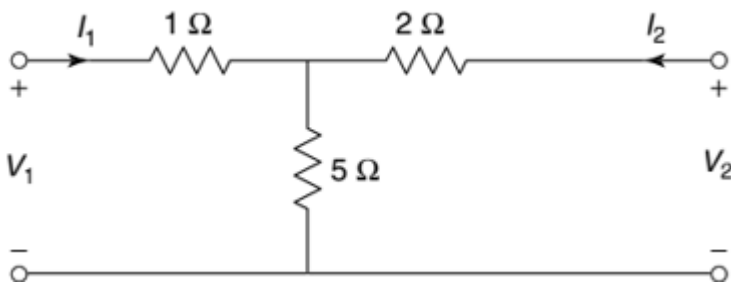
(4)

5B) Find Y-parameters for the network shown in Figure.



(4)

5C) Obtain ABCD parameters for the network shown in Figure.



(2)

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Event Name	Subject
058-END SEMSTER-ODD-2025	ECE_2122:NETWORK ANALYSIS

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	ECE_2122:NETWORK ANALYSIS

#	Level	No of Questions	Marks (%)
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1	Remember	0	0(0%)
2	Understand	1	2(4%)
3	Application	14	48(96%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)